

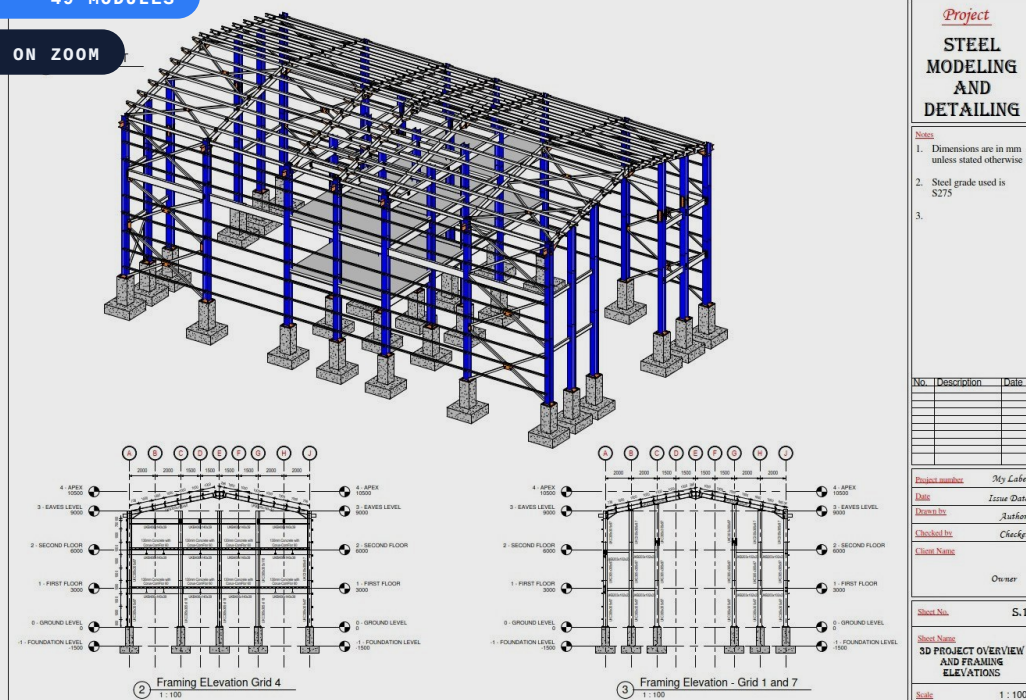
Build real, buildable structural models not just pretty renders.

2026
SEPT COHORT

SEPT COHORT

49 MODULES

100% LIVE ON ZOOM



STEEL PORTAL FRAME STRUCTURE 3D PROJECT OVERVIEW & FRAMING ELEVATIONS

FIG. 01

FEE
KSH 2,500 / TRACK

04 JOIN p. 8

01 THE PROGRAM

One program. Two *buildable* tracks.

Revit for Structures is a hands-on Autodesk Revit program for engineering students and early-career professionals who want to build real, buildable structural models. Two focused tracks Reinforced Concrete and Steel each run from first launch to finished documentation: modelling, reinforcement and connections, schedules, and sheets that print clean.

01

Model the whole frame

Columns to slabs,
rafters to bracing.

02

Detail like a pro

Rebar,
tags, bolts, plates & cuts.

03

Quantify & schedule

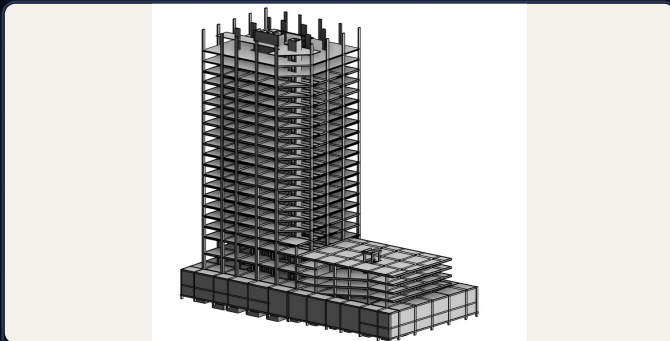
Bar bending schedules & takeoffs.

04

Document & deliver

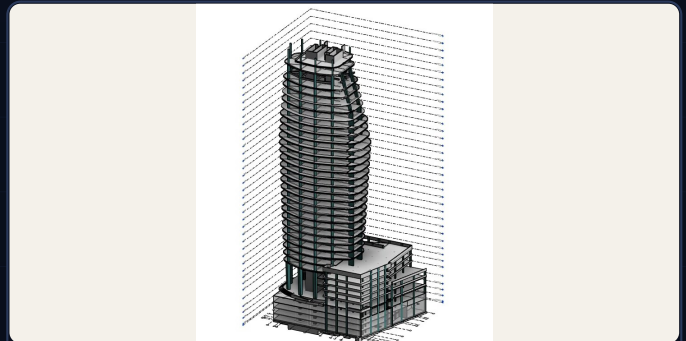
Sheets,
PDFs, CAD & analysis handoff.

WHAT YOU WILL MODEL REAL, STUDENT-GRADE OUTPUT



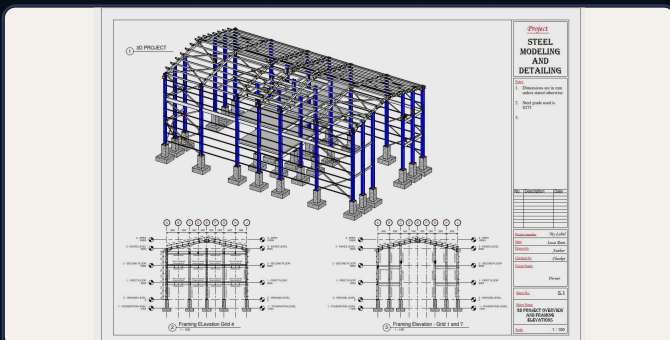
21 FLOORS + 2 BASEMENTS

FIG. 02



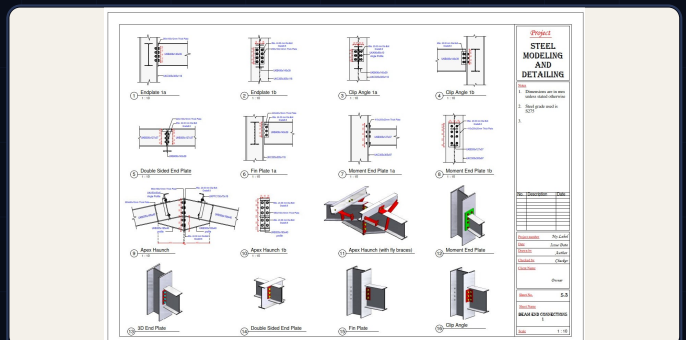
32 FLOORS + 4 BASEMENTS

FIG. 03



STEEL PORTAL FRAME STRUCTURE

FIG. 04



STEEL CONNECTIONS

FIG. 05

CHOOSE YOUR TRACK OR TAKE BOTH

TRACK 01

REINFORCED CONCRETE

20 modules Grids BBS p.3

TRACK 02

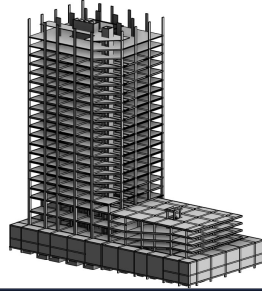
STEEL STRUCTURES

29 modules Columns Connections p.5

02 TRACK 01 REINFORCED CONCRETE 20 MODULES

FROM DATUMS TO DOCUMENTATION

Reinforced concrete, modelled to the rebar.



21 FLOORS + 2 BASEMENTS REINFORCED-CONCRETE TOWER FRAME, MODELLED IN REVIT

FIG. 02

A GROUNDWORK 01 05

01 BRIEF OVERVIEW OF REVIT

How Revit thinks Revit vs AutoCAD Why BIM wins

02 SETTING UP THE SOFTWARE

Essential libraries Autodesk extensions

03 BEGINNING WITH REVIT

New projects Templates Palettes tabs panels

04 DATUM ELEMENTS

Levels Grids Propagating extents Grid-bubble styling Linking CAD for grids

05 3D ELEMENTS, FIRST PASS

Foundations Columns Beams Slabs

B MODELLING THE FRAME 06 10

06 COLUMNS

Placement Loading families

07 FOUNDATIONS

Hosting on columns Strip footings Eccentricity Basement slabs Family libraries

08 BEAMS & SLABS

Copy level-to-level Slab creation Hollow-pot & waffle

09 CURVED BEAMS

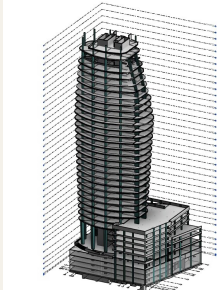
Modelling procedure, end to end

10 BEAM SYSTEMS & JUSTIFICATION

Beam-system modelling Layout rules Justification rules Deselecting cleanly

02 TRACK 01 REINFORCED CONCRETE, CONTINUED

Detail, reinforce, then **deliver**. *Stairs, rebar, tags, sheets.*



32 FLOORS + 4 BASEMENTS CURVED TOWER WITH LEVELS & GRIDS, MODELLED IN REVIT

FIG. 03

C DETAIL & REINFORCE 11 16

11 STAIRCASES & RAMPS

Stair types Architectural vs structural ramps Stairs modelled as ramps Shaft openings

12 VISIBILITY, GRAPHICS & FILTERS

Graphic overrides methods 1 4

13 ROOFS & TRUSSES

Sloped & gable-ended roofs System trusses Custom trusses

14 REINFORCEMENT

Sections first Foundations Columns Beams Slabs Stairs Ramps

15 ANNOTATIONS

Dimensions Spot elevations Spot coordinates Spot slopes

16 SECTION DETAILING

Breaklines Hatch regions Rebar tags Comments Element tags

D DOCUMENT & DELIVER 17 20

17 SCHEDULES & TAKEOFFS

Bar bending schedules Shape images in BBS Material takeoffs Sorting & filtering

18 DRAWINGS ON SHEETS

Titleblock sizes System blocks Custom blocks Text & fields Saving family templates Duplicates Placing views

19 PRINTING TO PDF

Sheet setup Single sheets Batch printing

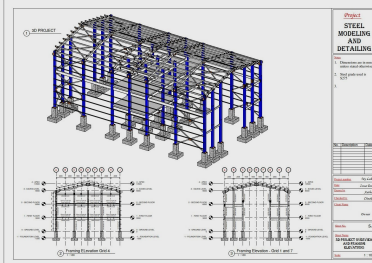
20 EXPORT & HANDOVER

Export options To AutoCAD To analysis

03 TRACK 02 STEEL STRUCTURES 29 MODULES

COLUMNS RAFTERS BEAMS BRACING

Steel, from first column to custom connection.



STEEL PORTAL FRAME STRUCTURE EAVES, APEX & FRAMING ELEVATIONS, MODELLED IN REVIT

FIG. 04

A GROUNDWORK 01 04

01 BRIEF OVERVIEW OF REVIT

How Revit thinks Revit vs CAD Why BIM wins

02 SETTING UP THE SOFTWARE

Essential libraries Autodesk extensions

03 BEGINNING WITH REVIT

New projects Templates Palettes tabs panels

04 DATUM ELEMENTS

Levels Grids Elevation markers Propagating extents Grid-bubble styling Linking CAD

B THE STEEL FRAME 05 10

05 STEEL COLUMNS

Placement Loading families

06 CONCRETE COLUMNS

Placement Loading families

07 FOUNDATIONS

Hosting on columns Extended family libraries

08 RAFTERS

Placement Steel beam sections Attaching columns Copying grid-to-grid

09 STEEL BEAMS

Sections Floor beams Justification rules Copying Eaves-level beams

10 BEAM SYSTEMS & JUSTIFICATION

Beam-system modelling Layout rules Justification rules Deselecting

Roofs, openings, bracing & views.

Composite decks to live sections.

C ROOF, BRACING & VIEWS 11 21

11 COMPOSITE BEAM & SLAB

Composite beams Composite slabs Cantilevered edges

12 FLOOR OPENINGS

The shaft-opening tool

13 CURVED BEAMS

Modelling procedure, end to end

14 SLOPED / SLANTED BEAMS

The offset method Framing elevations

15 PURLINS & SIDE RAILS

Reference-plane method Beam-system method Side rails

16 BRACINGS

Vertical bracing Horizontal bracing

17 VISIBILITY, GRAPHICS & FILTERS

Graphic overrides methods 1-4 Hiding elements

18 SECTIONS

Section types Cutting live sections

19 REINFORCEMENT

Cover setup Foundations Columns Slabs

20 SPLITTING ELEMENTS

Clean splits, buildable parts

21 CONNECTIONS FIRST PRINCIPLES

Uploading connections Applying them Propagating across the frame

NEXT THE CONNECTION LIBRARY

Thirteen connection families plus your own customs overleaf.

P. 07

The connection library.

Applied, propagated, detailed.

22 CONNECTION TYPES THE LIBRARY

Main types first, then every family you will meet on a real frame

Main types

Haunch

Apex

Base plate

Gusset bracing

Gusset column-base

Purlin / side-rail

End plate, single-sided (simple)

End plate, single-sided (moment)

End plate, double-sided (simple)

Fin plate

Clip angle

Column splice

Custom connections

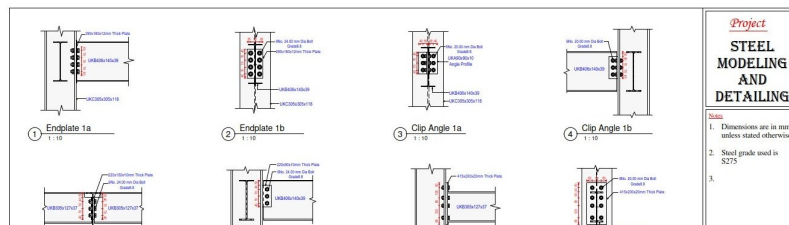
APPLIED LIVE

MODULE 21

UPLOAD

APPLY

PROPAGATE



STEEL CONNECTIONS END PLATES, CLIP ANGLES & MOMENT DETAILS AT 1:10

FIG. 05

D FABRICATION 23

23 FABRICATION & PARAMETRIC CUTS

Elements & modifiers Where cuts live Use & location

E DOCUMENT & DELIVER 24 29

24 ANNOTATIONS

Dimensions Spot elevations Coordinates Slopes Custom bolt & plate tags

25 SECTION DETAILING

Breaklines Hatch regions Rebar tags Comments Element tags

26 SCHEDULES & TAKEOFFS

Bar bending schedules Shape images in BBS Material takeoffs Sorting & filtering

27 DRAWINGS ON SHEETS

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Sheet setup Single sheets Batch printing

29 EXPORT & HANDOVER

Export options To AutoCAD To analysis

04 JOIN THE COHORT

Join the September cohort.

Live, online and hands-on from the third week of September.

SCHEDULE

START	Mon 14 Sept 2026
RHYTHM	Weekday evenings
FORMAT	Live on Zoom
TRACK	4 weeks each

Take either track or both, back to back.

INVESTMENT

KSH **2,500**

per track live instruction + materials

Students & early-career engineers

No prior Revit experience needed.

Just a laptop that runs Revit.

HOW IT WORKS

1 Reserve your seat

Write to us via the website and pick a track.

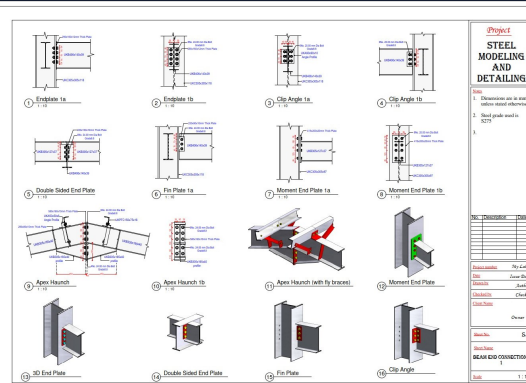
2 Set up Revit

We guide libraries & extensions before day one.

3 Build live

Evenings on Zoom model, detail, document.

GRADUATE WITH SHEETS LIKE THIS



STEEL CONNECTIONS BEAM END CONNECTIONS (S.3) AT 1:10, MODELLED & DETAILED IN REVIT

S.3

SEATS ARE LIMITED SEPTEMBER COHORT

structraining.co.ke

Reserve your track Concrete, Steel, or both.

ENROL

Autodesk Revit is a trademark of Autodesk, Inc. Structra Training is an independent program.